

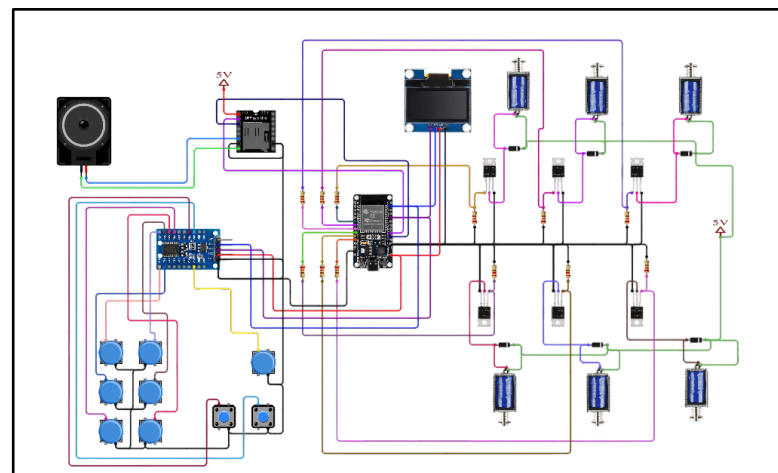
Design and Implementation of a Bidirectional Bangla Braille Translation System

OBJECTIVES

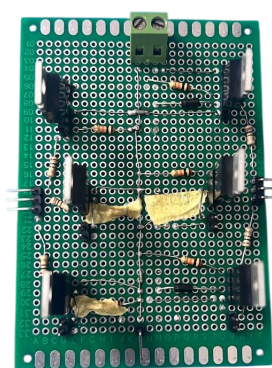
- Build a low cost, bidirectional Bangla Braille learning device.
- Convert Bangla text into tactile Braille using a refreshable cell.
- Convert Braille into Bangla text with display and audio feedback.

KEY FEATURES

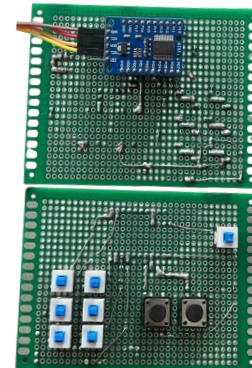
- Reading mode: 6-solenoid Braille cell raises dots letter by letter.
- Writing mode: 6-button input, shown on the OLED and spoken aloud.
- Wi-Fi upload: Bangla text is sent from a phone or laptop to the ESP32 through a locally hosted webpage.
- Proper Grammar: Maintains proper Bangla and Braille grammar.



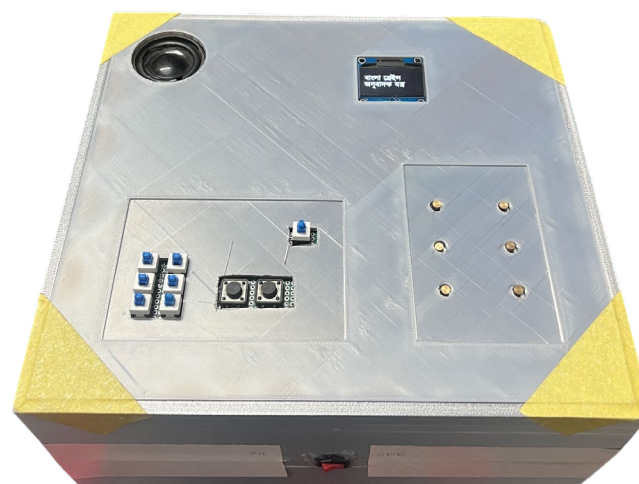
System Schematic



Solenoid Driver Circuit



Input Circuit



Hardware Implementation

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YouTube Link



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EEE 416 – Microprocessor and Embedded
Systems Laboratory

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